

Disseminated Cutaneous Leishmaniasis (DCL) in Immunosuppressed Patients

Immunosuppressed patients, particularly those with HIV infection, may develop disseminated cutaneous leishmaniasis (DCL) without species-specific association. DCL typically follows a classical cutaneous leishmaniasis (CL) lesion, with gradual development of non-ulcerated, parasite-laden nodules. These lesions have a predilection for exposed areas such as the extremities and face, where coalescence can result in leonine facies. Visceral involvement is absent. The disease follows a progressive or chronic course and is characterized by frequent relapse after treatment.

Diagnosis of Cutaneous Leishmaniasis

Due to the potential toxicity of anti-leishmanial therapies, confirmation of diagnosis is essential.

Even when combining smear, histology, and culture, the parasite may not be detected in 10% to 20% of cases. Diagnostic challenges are more pronounced in New World leishmaniasis, especially in chronic lesions (>6 months duration). Therefore, using multiple diagnostic methods is recommended for optimal detection.

Diagnostic Procedures

Skin Biopsy

A skin biopsy should be taken from the infiltrated margin of the lesion and divided into three parts for: - Impression smear - Histologic examination - Culture

Smears

Smears can be obtained via fine needle aspiration or tissue scrapings. Success rates range from 50% to 80%.

Procedure: - Air-dry the sample - Fix with methyl alcohol - Stain with Giemsa
- Examine under oil immersion

Impression smears—prepared by gently pressing the biopsy specimen against a glass slide 2–5 times—offer higher sensitivity than routine histologic sections.

Histopathology

In both Old World (OW) and New World (NW) CL, early lesions show a dense, diffuse mixed inflammatory infiltrate composed mainly of histiocytes, with scattered multinucleated giant cells, lymphocytes, and plasma cells. Russell bodies (intracytoplasmic eosinophilic immunoglobulin deposits) may be present.

Features less commonly observed: - Preservation of the papillary dermis (grenz zone) - Presence of eosinophils and neutrophils - Perineural inflammation

The hallmark of leishmaniasis—seen in approximately 70% of cases—is the presence of numerous **amastigotes** (Leishman-Donovan bodies) within histiocytes and extracellularly.

- **Giemsa stain:** stains the parasite non-metachromatically; kinetoplast appears bright red.

As lesions evolve, amastigote numbers decrease, and histology resembles chronic cutaneous leishmaniasis (CCL), characterized by nodular or diffuse noncaseating tuberculoid granulomatous dermatitis.

Other histologic findings: - Variable epidermal hyperplasia and ulceration - Scarring with marked loss of elastic fibers

In **DCL**, the infiltrate is diffuse, composed of vacuolated macrophages containing abundant intra- and extracellular amastigotes. The main differential diagnosis is **lepromatous leprosy**.

Histologic Differential Diagnosis (diseases with parasitized macrophages): -

Histoplasmosis: yeasts with surrounding halo - **Granuloma inguinale:** safety-pin-like Donovan bodies - **Rhinoscleroma:** Mikulicz cells - Other considerations: blastomycosis, paracoccidioidomycosis, toxoplasmosis, trypanosomiasis

Other Diagnostic Procedures

Laboratory animal inoculation (xenodiagnosis) may be useful in cases with low parasite load when conventional methods fail.